

# Weather App Project Report

Course Project, Project option: 2

## 1. What I built

I chose option 2 and built weather app with HTML, CSS and JavaScript. It fulfills the fundamental functions of a weather forecast application, allowing users to look up the weather for any city or use GPS to check the forecast for their current location.

On the homepage, users can not only check the weather but also add cities to their favorites—data that persists across page refreshes or reloads—and toggle temperature units between °C, °F, and K. I used both Open-Meteo and MET Norway, two websites at the same time, so the app pulls from two data sources. Therefore I draw the temperature comparison chart for the next 24 hours. Users can then check the weather for the next 24 hours, with hourly conditions displayed (data sourced from Open-Meteo); scrolling down reveals the forecast for the upcoming seven days.

## 2. Tools used

HTML5, CSS3 and JavaScript without react or framework as required. Browser APIs such as `fetch()`, `LocalStorage`, `<Canvas>` for the chart, and `navigator.geolocation.getCurrentPosition()` etc. I used Gemini flash3.6 to help me review how to create the background color gradient effect and a feature to prevent duplicate saves, as well as issues regarding code errors and debugging. External data sources were utilized—Open-Meteo ([api.open-meteo.com](https://api.open-meteo.com)) as the primary source and MET Norway ([api.met.no](https://api.met.no)) as the secondary source to generate a comparative line chart.

## 3. Points I think project deserves

| Feature                    | Points | Why                                       |
|----------------------------|--------|-------------------------------------------|
| PDF report                 | 3      | This document                             |
| Responsive(desktop+mobile) | 4      | Tested on my phone                        |
| Favorites                  | 3      | Canvas line chart, temperature comparison |
| °C/°F/K switch             | 2      | Call <code>showWeather()</code> again     |
| Works on                   | 3      | Tested on all these 4 websites            |

|                                            |    |                                                                                                                           |
|--------------------------------------------|----|---------------------------------------------------------------------------------------------------------------------------|
| Firfox,Safari,Edge and Chrome              |    |                                                                                                                           |
| Clear dieectory                            | 2  | Files separated                                                                                                           |
| Location search                            | 1  | Open-Meteo API                                                                                                            |
| searchCity                                 | 2  | Call the Open-Meteo geocoding API and pass the data back to getWeather()                                                  |
| GPS location                               | 2  | Navigator.geolocation                                                                                                     |
| 2 data sources                             | 3  | Open-Meteo and MET Norway                                                                                                 |
| Current weather                            | 1  | Display                                                                                                                   |
| 24h forecast                               | 2  | Generate cards over next 24 hours                                                                                         |
| 7day forecast                              | 3  | Iterate through the daily array,                                                                                          |
| Icons                                      | 3  | Icons shown up                                                                                                            |
| Background color change                    | 2  | Background color changes with temperature                                                                                 |
| Duplicate prevention<br>Persistent storage | 3  | The same city will not be saved to favorites more than once, favorites list is not cleared every time the page refreshes. |
| Total                                      | 39 |                                                                                                                           |

**3. No AI assistance or tools were used in this assignment.**